

Breast Cancer Classification Using Machine Learning: Classification Models, Evaluation Metrics & Handling Imbalanced Data

1. Objective

The objective of this project was to develop a binary classification system using the Breast Cancer Dataset and evaluate the performance of multiple machine learning classification algorithms. The project focused on feature preprocessing, model training, performance evaluation using classification metrics, ROC-AUC analysis, handling class imbalance, and selecting the most reliable classification model.

2. Dataset Description

The Breast Cancer Dataset was obtained from the Scikit-Learn library. The target variable represents whether a tumor is **Malignant (0)** or **Benign (1)**. The input features consist of 30 numerical attributes computed from digitized images of breast mass samples, including radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension. The dataset contains **569 samples**, **30 input features**, and **2 target classes**.

3. Methodology

Data Preparation

The Breast Cancer Dataset was loaded using Scikit-Learn and converted into a pandas DataFrame. The input features (**X**) and target labels (**y**) were separated before model training.

Train-Test Split

The dataset was divided into training and testing sets using an **80:20 ratio** with a **random state of 42**.

Stratified sampling (**stratify = y**) was applied to preserve the class distribution in both training and testing datasets.

Feature Scaling

Since Logistic Regression performs better when features are on a similar scale, **StandardScaler** was applied to standardize the training and testing data.

Machine Learning Models

Two classification algorithms were trained and evaluated:

1. Logistic Regression
2. Decision Tree Classifier

Handling Class Imbalance

A second Logistic Regression model was trained using **class_weight = “balanced”** to demonstrate one of the standard techniques used for handling imbalanced datasets. Although the Breast Cancer Dataset is relatively balanced, this technique is widely used in real-world classification problems.

4. Evaluation Metrics

Model performance was evaluated using the following classification metrics:

- Confusion Matrix
- Precision
- Recall
- F1-Score
- ROC Curve
- Area Under the ROC Curve (ROC-AUC)

The Confusion Matrix provides detailed information about correct and incorrect predictions. Precision measures prediction accuracy for positive cases, Recall measures the ability to correctly identify positive cases, and the F1-Score provides a balance between Precision and Recall. ROC-AUC evaluates the model’s ability to distinguish between the two classes across different decision thresholds.

5. Results and Performance Comparison

Model	AUC Score
Logistic Regression	0.995370
Decision Tree Classifier	0.915675

6. Best Model Selection

Among the evaluated models, **Logistic Regression** achieved the highest **ROC-AUC Score (0.995370)**, demonstrating excellent classification performance and superior discrimination between malignant and benign tumors. The Decision Tree Classifier also produced good results with an AUC Score of **0.915675**, but it was comparatively less accurate and more susceptible to overfitting. Therefore, Logistic Regression was selected as the best-performing classifier for this dataset.

7. Model Validation

The model was evaluated using a Confusion Matrix, Classification Report, ROC Curve, and AUC Score. The ROC curve was positioned close to the upper-left corner of the graph, indicating excellent classification capability. The high AUC Score confirmed that the Logistic Regression classifier successfully distinguished between malignant and benign tumors with very high reliability.

Additionally, the balanced Logistic Regression model demonstrated how **class_weight = “balanced”** can be used to improve learning when datasets contain unequal class distributions. Since the Breast Cancer

Dataset is already relatively balanced, only minor differences were observed between the baseline and balanced models.

8. Conclusion

This project demonstrated the complete machine learning workflow for binary classification, including data preprocessing, feature scaling, model training, evaluation using multiple classification metrics, ROC-AUC analysis, handling class imbalance, and model comparison.

Two classification algorithms were evaluated on the Breast Cancer Dataset. Based on ROC-AUC performance, **Logistic Regression** achieved the best overall classification results with an **AUC Score of 0.995370**, making it the preferred model for this task.

The project highlights the importance of selecting appropriate evaluation metrics instead of relying solely on accuracy when solving classification problems. Future improvements could include hyperparameter tuning, Random Forest, Support Vector Machines (SVM), Gradient Boosting, and XGBoost classifiers for further performance enhancement.

This report mirrors the **structure, tone, and formatting** of your previous Task 2 and Task 3 reports, giving all three internship submissions a consistent and professional appearance.